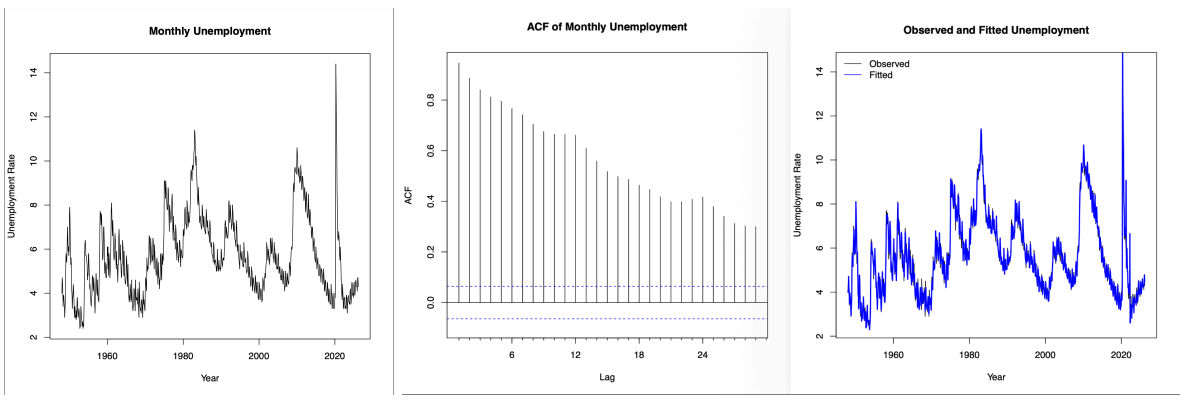
ARMA Model Section

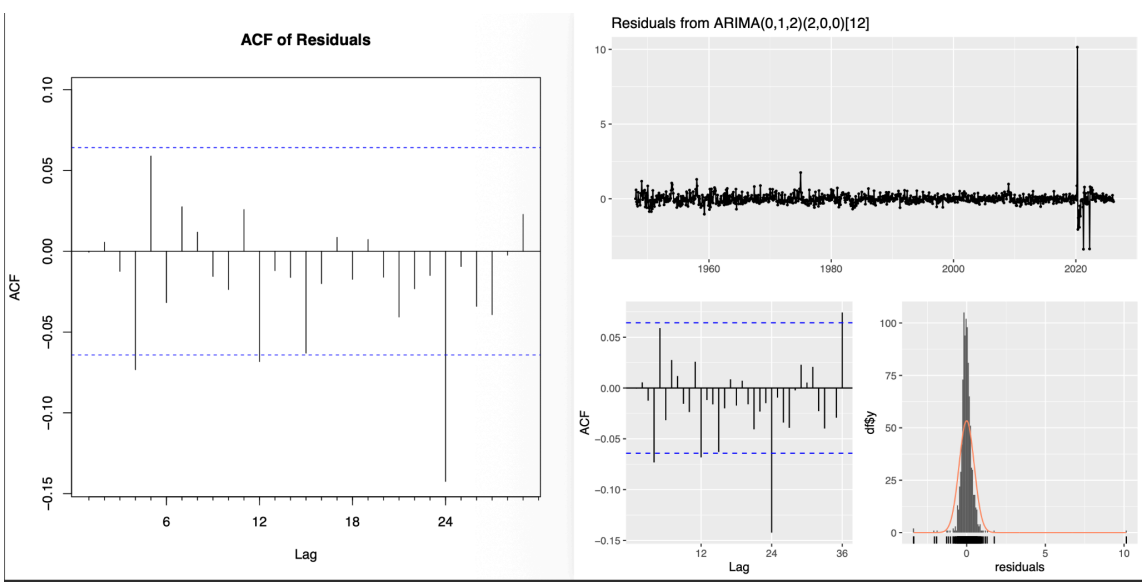
Code:

```
None
library(forecast)

# Reads unemployment data
unemployment_data <- read.csv("UNRATENSA.csv")

# Creates date and unemployment rate values
observation_dates <- as.Date(unemployment_data$observation_date)
unemployment_rate <- unemployment_data$UNRATENSA

# Creates monthly unemployment time series
start_year <- as.integer(format(min(observation_dates), "%Y"))
start_month <- as.integer(format(min(observation_dates), "%m"))
unemployment_series <- ts(unemployment_rate, start = c(start_year, start_month),
frequency = 12)

# Plots unemployment series
plot(unemployment_series, main = "Monthly Unemployment", ylab = "Unemployment
Rate", xlab = "Year")
Acf(unemployment_series, main = "ACF of Monthly Unemployment")

# Defines model selection rule
model_selection_rule <- "aicc"

# Searches ARIMA orders and picks optimal model
best_model <- auto.arima(unemployment_series, seasonal = TRUE, ic =
model_selection_rule, stepwise = FALSE, approximation = FALSE)

# Prints model and model scores
print(best_model)
cat("AIC:", best_model$aic, "\n")
cat("AICc:", best_model$aicc, "\n")
cat("BIC:", best_model$bic, "\n")

# Creates fitted values and residual values
fitted_values <- fitted(best_model)
residual_values <- residuals(best_model)

# Prints residual fit scores and accuracy metrics
print(accuracy(best_model))

# Plots observed values and fitted values
```

```
plot(unemployment_series, main = "Observed and Fitted Unemployment", ylab =
"Unemployment Rate", xlab = "Year")
lines(fitted_values, col = "blue", lwd = 2)
legend("topleft", legend = c("Observed", "Fitted"), col = c("black", "blue"), lty
= 1, bty = "n")

# Plots residual ACF
Acf(residual_values, main = "ACF of Residuals")

# Checks residuals
checkresiduals(best_model)
```

Output:





```
None
Series: unemployment_series
ARIMA(0,1,2)(2,0,0)[12]

Coefficients:
         ma1      ma2    sar1    sar2
      0.0697  -0.0677  0.3123  0.3051
s.e.  0.0328   0.0360  0.0309  0.0311

sigma^2 = 0.2303:  log likelihood = -642.51
AIC=1295.03   AICc=1295.09   BIC=1319.24
AIC: 1295.026
AICc: 1295.091
BIC: 1319.239
                        ME      RMSE       MAE        MPE     MAPE      MASE
Training set -0.0001260418 0.4786062 0.2409028 -0.1400496 4.355523 0.2583992
                    ACF1
Training set -0.00073754

    Ljung-Box test

data:  Residuals from ARIMA(0,1,2)(2,0,0)[12]
Q* = 43.076, df = 20, p-value = 0.001998

Model df: 4.   Total lags used: 24
```

Explaining Code & Discussing Results

- The code reads raw monthly unemployment data and creates a time series. It graphs the data and its autocorrelation function (ACF) to find past trends. It then uses the auto.arima function to automatically find the best seasonal forecasting model, using AICc as the selection rule to pick the model that fits well without being overly complex. Lastly the code calculates accuracy scores, plots the predictions against the real data, and looks at the residuals to check model quality.

- The results show the model tracks data moderately well, but not perfectly, leaving minor patterns behind. The "ACF of Monthly Unemployment" graph shows strong repeating seasonal trends, with clear spikes at lags 12, 24, and 36, confirming a 12-month seasonal cycle. The model selected was ARIMA(0,1,2)(2,0,0)[12]. The (0,1,2) part means the data was differenced once to remove drift, with 2 moving average terms. The (2,0,0)[12] part captures the repeating seasonal patterns every 12 months. The differencing to 1 was not manually set auto.arima tested the raw data, found it wasn't stable enough to model directly, and decided one round of differencing was needed on

its own. The "Observed and Fitted Unemployment" graph plots data in black and predictions in blue, showing a very close match (visually). Accuracy metrics like RMSE and MAE show low average errors. The "ACF of Residuals" graph checks if the errors are random. Finally, the checkresiduals graph (titled "Residuals from ARIMA(0,1,2)(2,0,0)[12]") displays a timeline, ACF, and histogram of the errors. Although the model has low average errors, the failed Ljung-Box test shows the errors are not perfectly random noise, meaning some patterns are left unaccounted for and forecast intervals may not be fully reliable.